

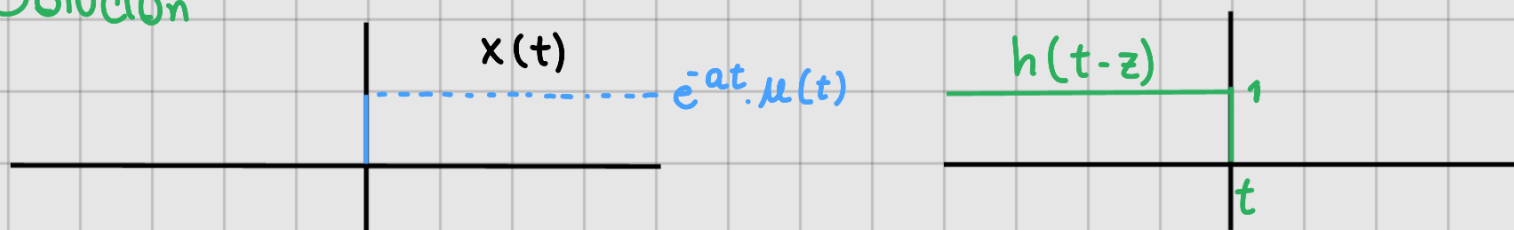
Convolución Parte 2

Ejercicio Repaso

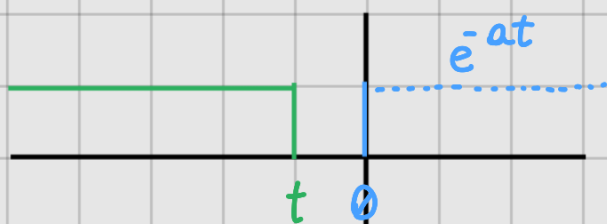
$$x(t) = e^{-at} \cdot \mu(t) \quad a > 0$$

$$h(t) = \mu(t)$$

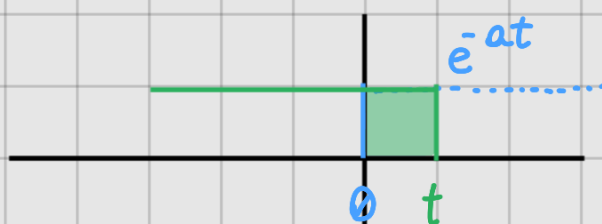
Solución



Convolución

Para $t < 0$ 

No hay convolución

Para $t > 0$ 

$$y(t) = \int_0^t e^{-az} \cdot 1 \cdot dz$$

$$y(t) = -\frac{1}{a} \cdot e^{-az} \Big|_0^t$$

$$y(t) = -\frac{1}{a} \cdot e^{-at} - \left(-\frac{1}{a} \cdot 1 \right)$$

Propiedades de la convolución

- Conmutativa : $x(t)$ puede actuar como $h(t)$ (Reflejar y liberar)
 $h(t)$ puede actuar como $x(t)$ (Señal fija en eje)
- Asociativa : Si tuviera que hacer convolución de 3 señales no importa el orden.

$$[x(t) * h_1(t)] * h_2(t) = x(t) * [h_1(t) * h_2(t)]$$

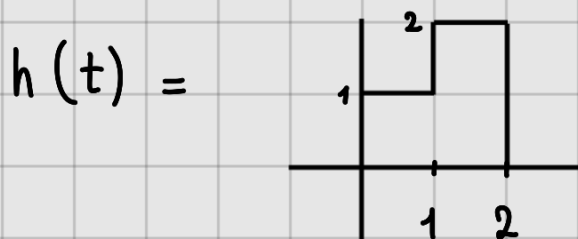
- Distributiva respecto de la suma ó resta :

$$x(t) * [h_1(t) + h_2(t)] = [x(t) * h_1(t)] + [x(t) * h_2(t)]$$

- Elemento neutro $\delta(t)$: La convolución de una señal con un impulso es el valor de la señal donde estaba el impulso.
 $y(t) = x(t)$

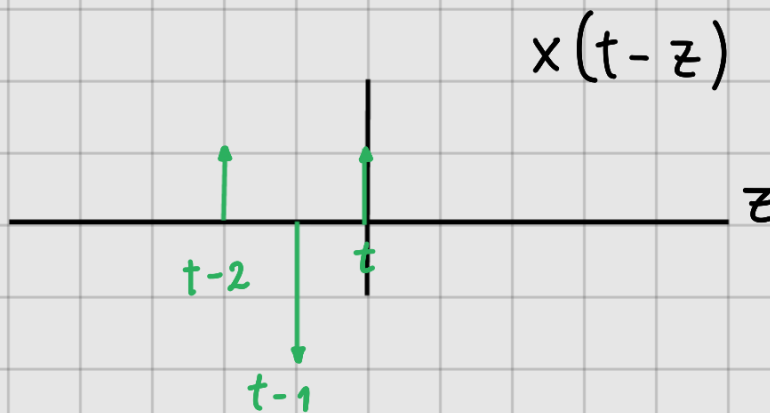
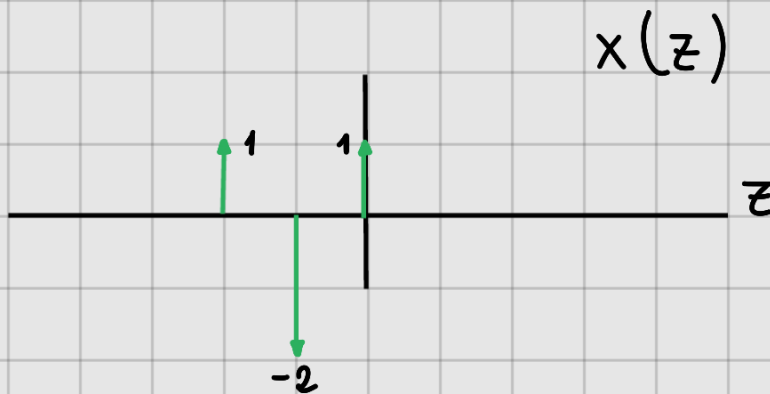
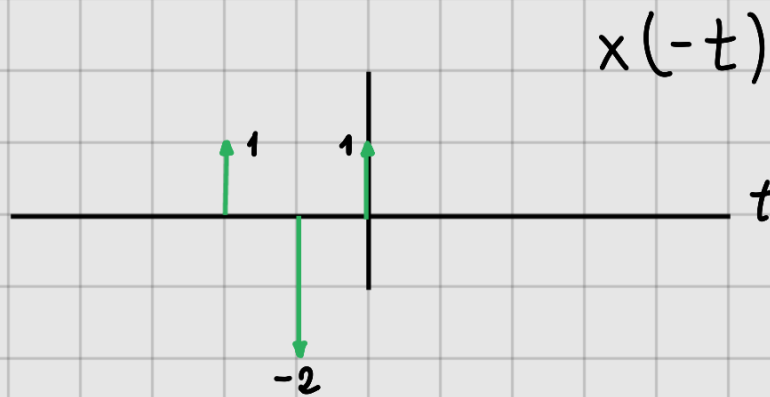
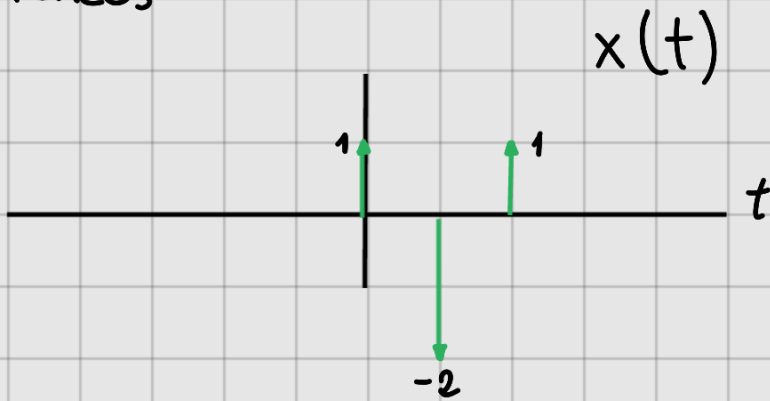
Ejemplo:

$$x(t) = \delta(t) - 2\delta(t-1) + \delta(t-2)$$



Tomamos $x(t)$ como la señal que se refleja y se libera, porque es más fácil simplemente

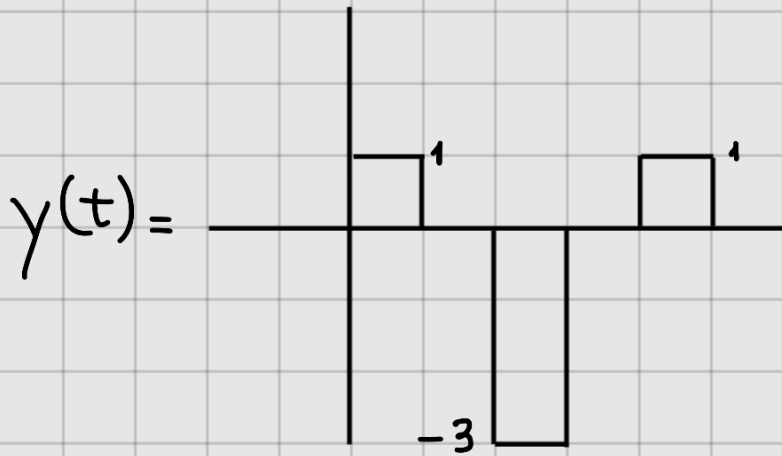
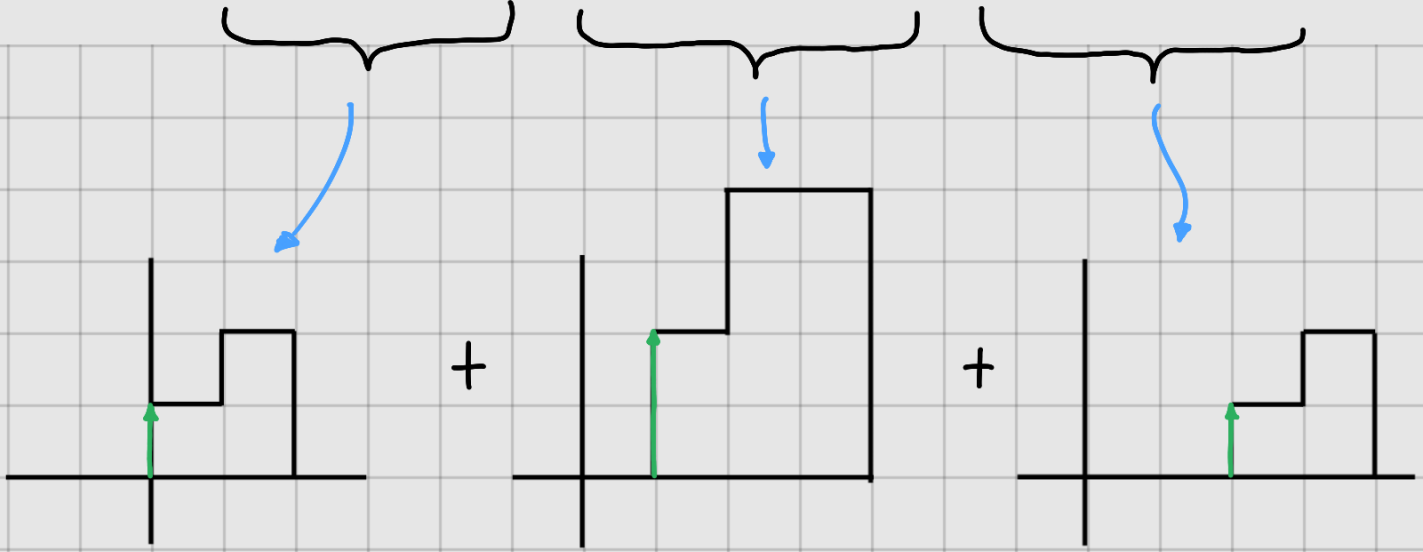
Entonces



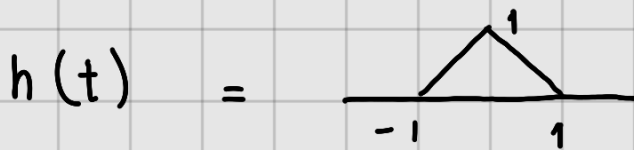
$$y(t) = h(t) * \{ \delta(t) - 2\delta(t-1) + \delta(t-2) \}$$

Por propiedades

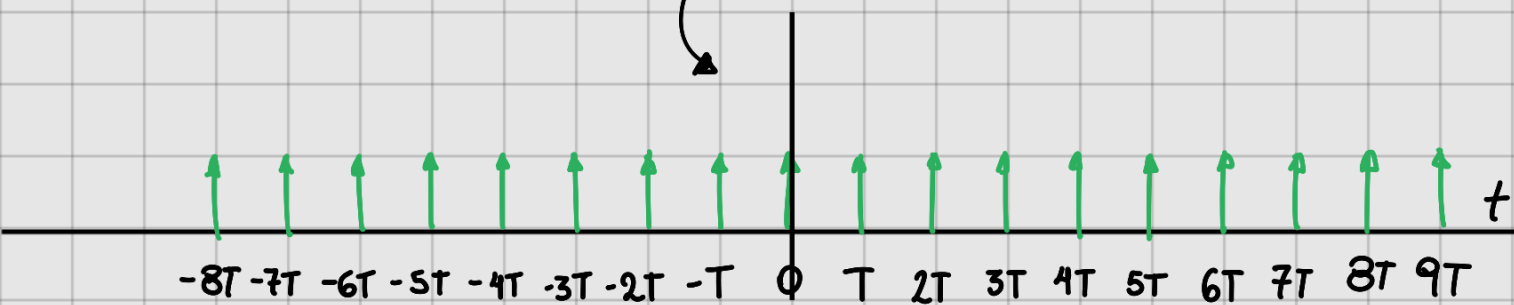
$$y(t) = h(t) * \delta(t) + h(t) \cdot -2\delta(t-1) + h(t) \cdot \delta(t-2)$$



2)

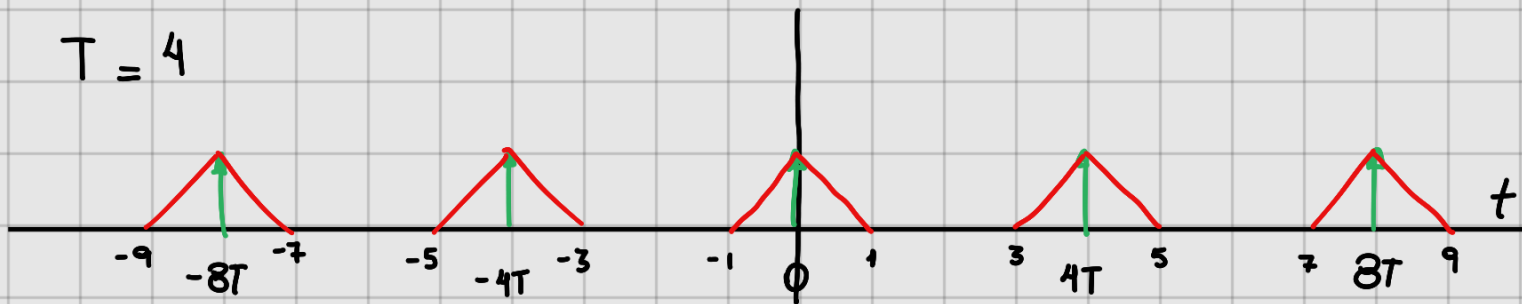


$$x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$$

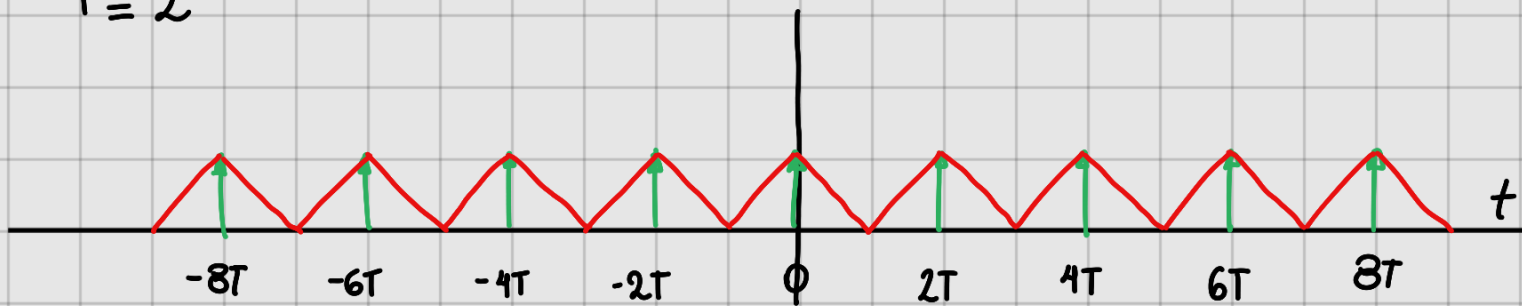


Analice si

$$T = 4$$



$$T = 2$$



$$T = 1$$

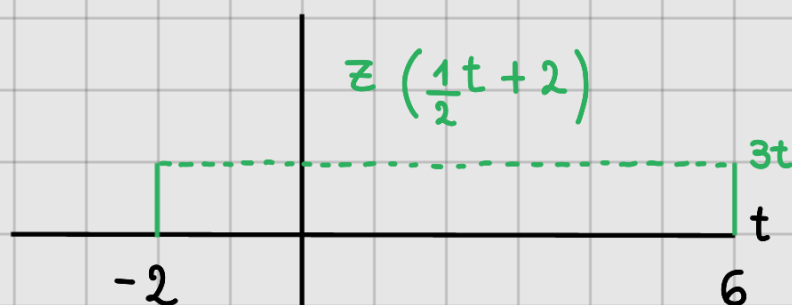
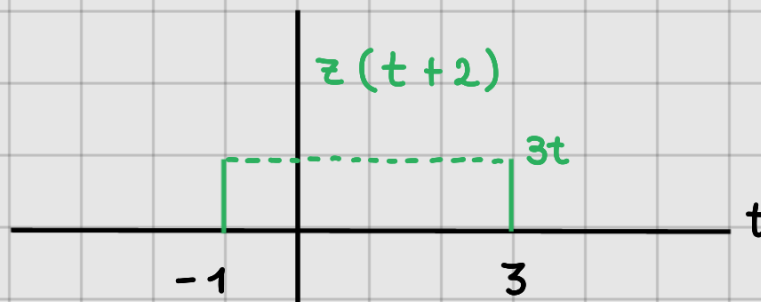
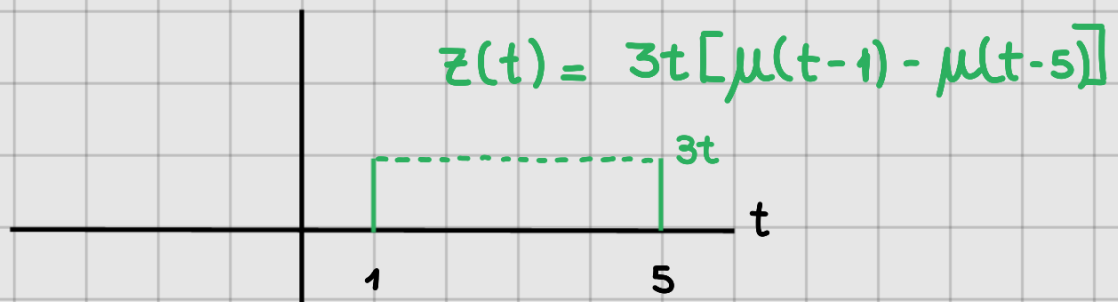
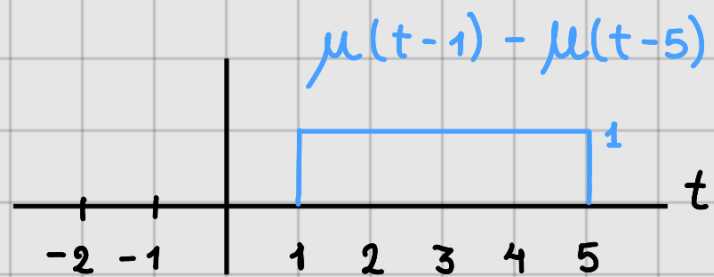
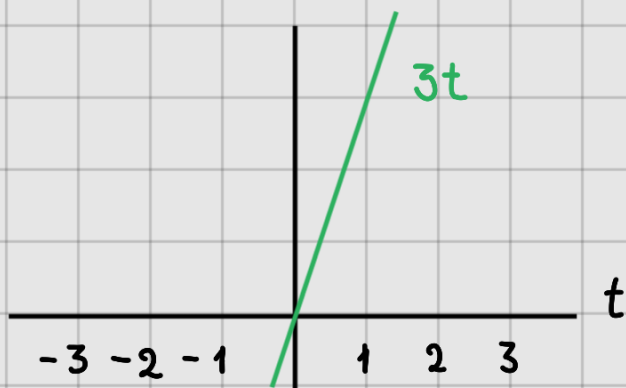


EJERCICIOS TIPO PARCIAL

1) $z(t) = 3t \cdot [\mu(t-1) - \mu(t-5)]$

Encontrar $z(0,5t+2)$

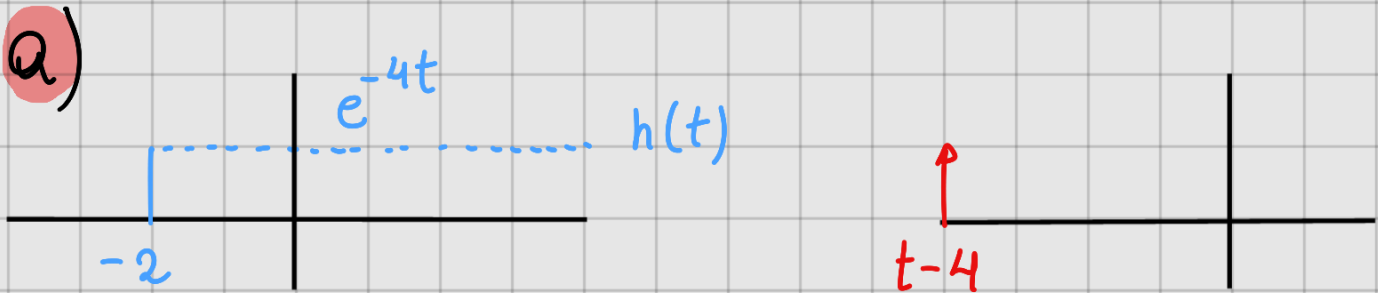
Solución



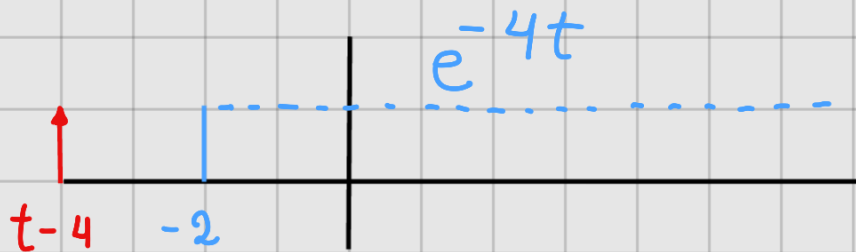
2) $h(t) = e^{-4t} \cdot \mu(t+2)$

$x(t) \begin{cases} a) \delta(t-4) \\ b) \mu(t+3) \end{cases}$

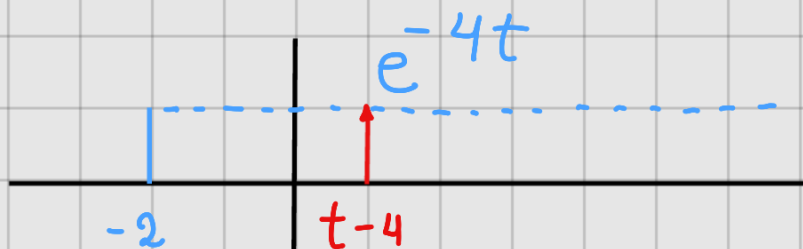
Solución



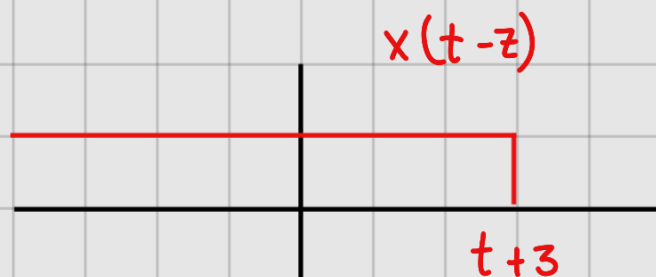
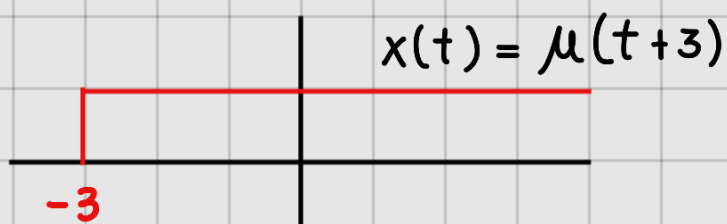
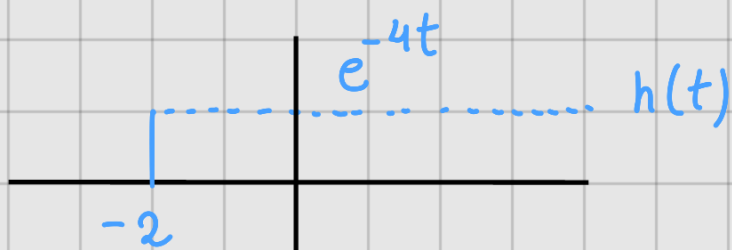
• Para $t-4 < 2 \longrightarrow t < 6 \quad y(t) = 0$



• Para $t-4 > 2 \longrightarrow t > 6 \quad y(t) = e^{-4(t-4)} \cdot \mu(t-4+2)$



b)



Convolución

• Para $t+3 < -2$

↓

$t < -5$

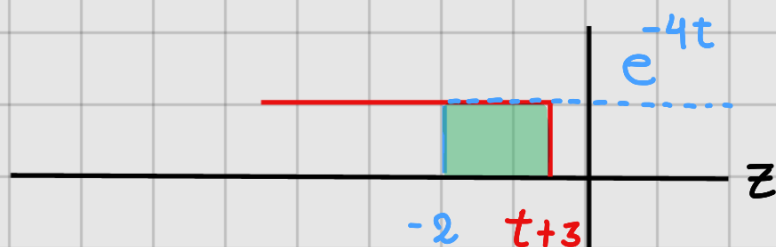


Integral de convolución

$y(t) = 0$ No hay convolución

• Para $t+3 > -2$

$$\boxed{t > -5}$$



Integral de convolución

$$y(t) = \int_{-2}^{t+3} e^{-4t} \cdot 1 \cdot dz$$

$$y(t) = -\frac{1}{4} \cdot e^{-4z} \Big|_{-2}^{t+3}$$

$$\boxed{y(t) = -\frac{e^{-4(t+3)}}{4} - \frac{e^8}{4}}$$

Dados :

$$x(t) = \mu(t+1) - \mu(t-2) + e^{-t} \cdot \mu(t-4)$$

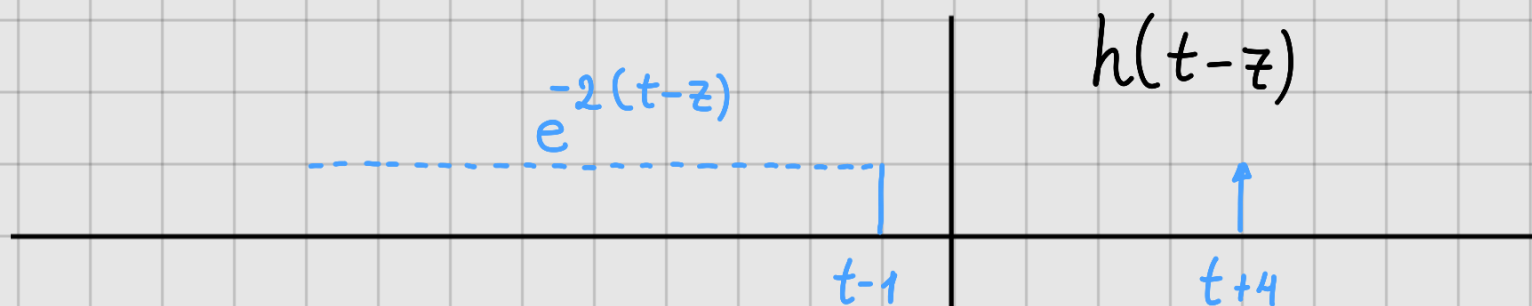
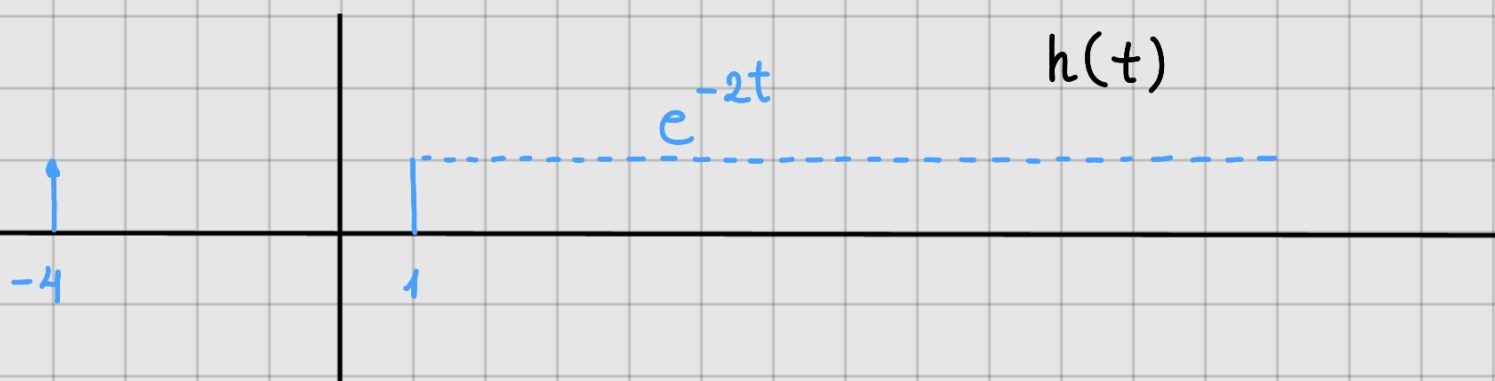
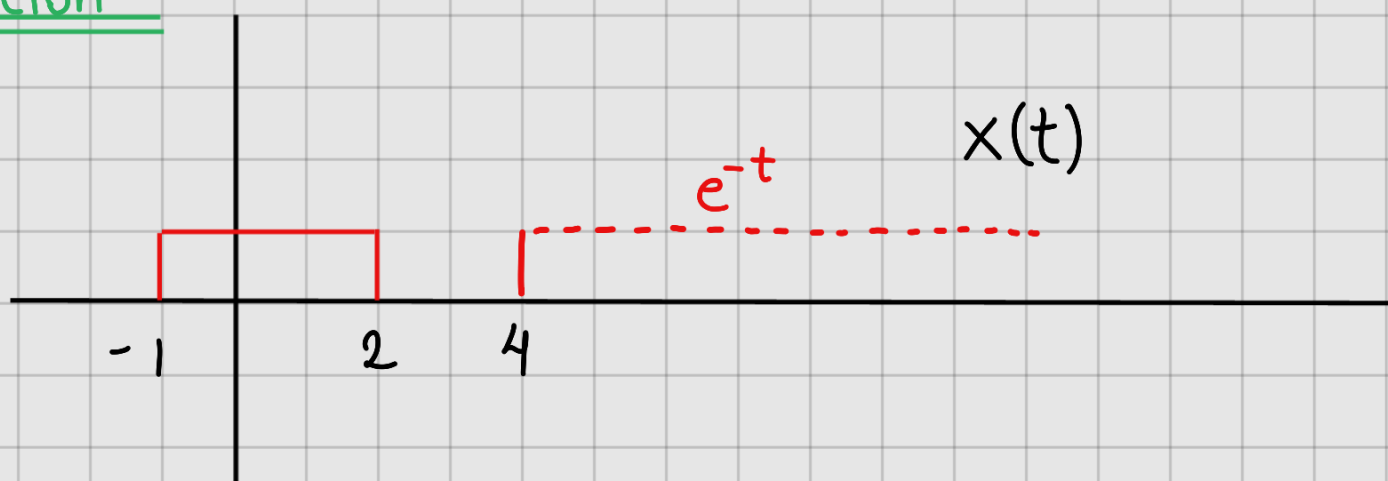
$$h(t) = e^{-2t} \cdot \mu(t-1) + \delta(t+4)$$

Calcular su salida LTI y hallar los valores para :

$$y(-3) = ?$$

$$y(0,5) = ?$$

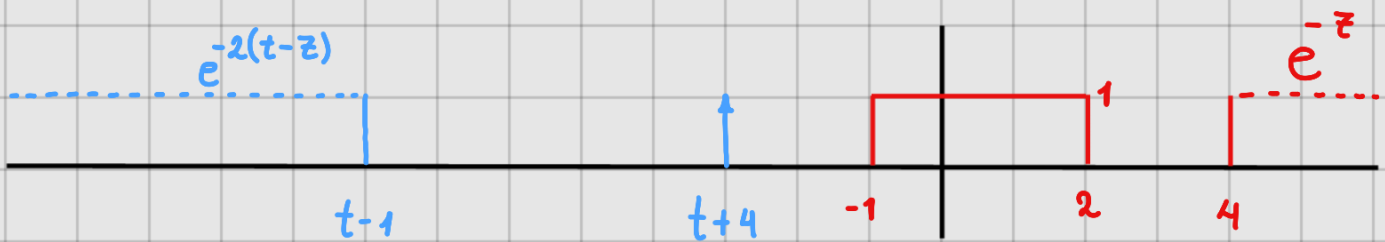
Solución



Convolución

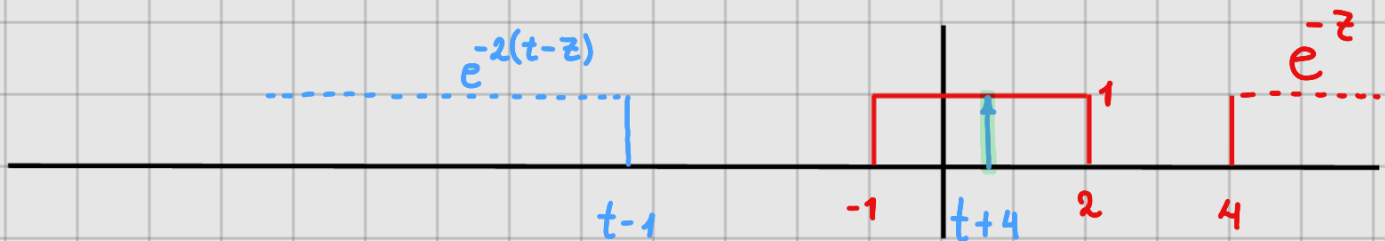
$$-\infty < t < -5 \longrightarrow y(t) = 0$$

Grafica de convolución:



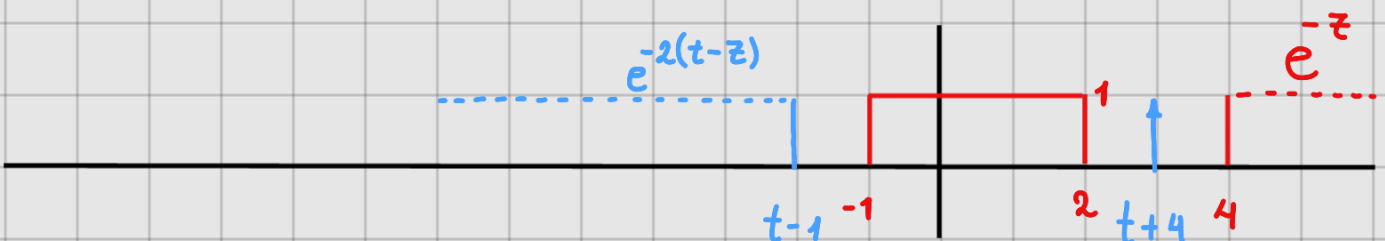
$$-5 < t < -2 \longrightarrow y(t) = 1 \cdot \delta(t+4) \longrightarrow 1$$

Grafica de convolución:



$$-2 < t < 0 \longrightarrow y(t) = 0$$

Grafica de convolución:



$$0 < t < 3 \rightarrow y(t) = \int_{-1}^{t-1} 1 \cdot e^{-2(t-z)} dz + e^{-z} \delta(t+4)$$

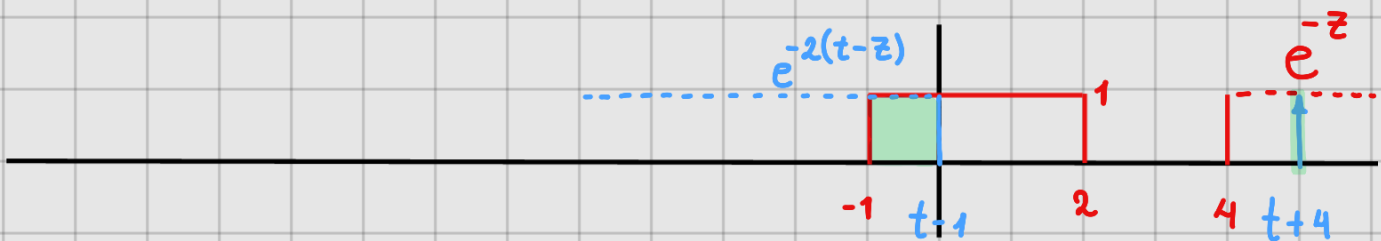
$$y(t) = \int_{-1}^{t-1} e^{-2t+2z} dz + e^{-(t+4)}$$

$$y(t) = \int_{-1}^{t-1} e^{-2t} \cdot e^{2z} dz + e^{-(t+4)}$$

$$y(t) = e^{-2t} \cdot \frac{1}{2} e^{2z} \Big|_{-1}^{t-1}$$

$$y(t) = \frac{e^{-2t}}{2} \cdot (e^{2(t-1)} - e^{2(-1)})$$

Grafica de convolución:

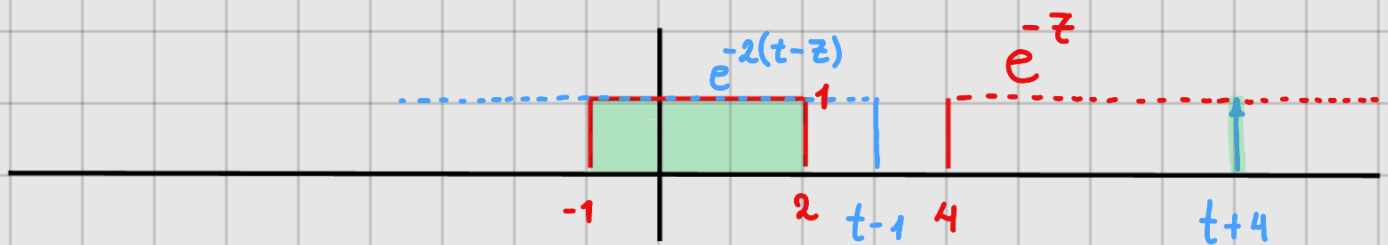


$$3 < t < 5 \rightarrow y(t) = \int_{-1}^2 1 \cdot e^{-2(t-z)} dz + e^{-z} \cdot \delta(t+4)$$

$$y(t) = \frac{e^{-2t}}{2} \cdot e^{2z} \Big|_{-1}^2 + e^{-(t+4)}$$

$$y(t) = \frac{e^{-2t}}{2} \cdot (e^{2(2)} - e^{2 \cdot (-1)}) + e^{-(t+4)}$$

Grafica de convolución



$$5 < t < \infty \rightarrow y(t) = \int_{-1}^2 1 \cdot e^{-2(t-z)} dz + \int_4^{t-1} e^{-z} \cdot e^{-2(t-z)} dz + e^{-z} \cdot \delta(t+4)$$

$$y(t) = \frac{e^{-2t}}{2} \cdot (e^{2(2)} - e^{2 \cdot (-1)}) + \int_4^{t-1} e^{-z} \cdot e^{-2t} \cdot e^{2z} dz + e^{-(t+4)}$$

$$y(t) = \frac{e^{-2t}}{2} \cdot (e^{2(2)} - e^{2 \cdot (-1)}) + e^{-2t} \cdot \int_4^{t-1} e^z \cdot dz + e^{-(t+4)}$$

$$y(t) = \frac{e^{-2t}}{2} \cdot (e^{2(2)} - e^{2 \cdot (-1)}) + e^{-2t} \cdot e^z \Big|_{-1}^{t-1} + e^{-(t+4)}$$

$$y(t) = \frac{e^{-2t}}{2} \cdot (e^{2(2)} - e^{2 \cdot (-1)}) + e^{-2t} \cdot (e^{(t-1)} - e^4) + e^{-(t+4)}$$

